

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)		
Student Name:	P. Saibhargav	Reg. No.:	192411049
Section / Batch:	2024/CSE.	Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:			100 Marks

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I P. Saibhargav (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Co1 AT1

P. Saibhargav

192411049

csA0618

① online video streaming platform
Given recurrence.

$$T(n) = 2T(n/2) + n$$

master theorem

general form:

$$T(n) = aT(n/b) + f(n)$$

$$a = 2 \quad b = 2 \quad f(n) = n$$

Calculate:

$$n \log_b a = n \log_2 2 = n$$

comparison:

$$f(n) = n = \Theta(n \log_2 2)$$

Therefore, case 2 of the master theorem applies.

$$T(n) = \Theta(n \log n)$$

② social media feed processing

Given

$$T(n) = T(n-1) + n$$

substitution method

$$T(n) = T(n-1) + n$$

substitute $T(n-1)$:

$$T(n) = T(n-2) + (n-1) + n$$

again:

$$T(n) = T(n-3) + (n-2) + (n-1) + n$$

After k expansions:

$$T(n) = T(n-k) + (n-k+1) + \dots + (n-1) + n$$

When $k=n-1$:

$$T(n) = T(1) + 2 + 3 + \dots + n$$

using:

$$1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

$$T(n) = \Theta(n^2)$$

$\dots \Theta(n^2)$ //

③ E-commerce Recommendation system.

$\Theta(n \log n)$

and

$\Theta(n^2)$

growth rate

comparison

input size

$\Theta(n \log n)$

$\Theta(n^2)$

small

fast

fast

medium

efficient

slower

large

much more efficient

very slow

for example, if $n = 1000$:

$$n \log_2 n \approx 1000 \times 10 = 10,000$$

$$n^2 = 1,000,000$$

Thus:

$$O(n \log n) \ll O(n^2)$$

$$= O(n \log n)$$

④ cloud storage processing system.

Given:

$$T(n) = 3T(n/2) + n$$

master theorem

general form:

$$T(n) = aT(n/b) + f(n)$$

Parameters:

$$a = 3$$

$$b = 2$$

$$f(n) = n$$

calculate:

$$n \log_2 a = n \log_2 3$$

since:

$$\log_2 3 \approx 1.585$$

we get:

$$n \log_2 3 = n^{1.585}$$

compare: $f(n) = n$.

with: $n^{1.585}$

Since: $n = O(n^{1.585 - \epsilon})$

$\epsilon > 0$, case 1 applies.

$$T(n) = \Theta(n \log^2 n)$$

or approximately:

$$T(n) = \Theta(n^{1.585})$$

⑤ Financial Analytics system.

Given: $T(n) = 2T(n/2) + n \log n$

Parameters: $a=2$ $b=2$ $f(n) = n \log n$

calculate:

$$n \log_b a = n \log_2 2 = n$$

Now compare:

$$f(n) = n \log n$$

with:

$$n \log_b a = n.$$

we have: $f(n) = \Theta(n \log n)$

$$T(n) = \Theta(n \log n \log n)$$

Thus:

$$T(n) = \Theta(n \log^2 n)$$